

Scenario Based - Set Qn 1

Scenario 1: Employee Salary Analysis

Question:

Find the names and salaries of employees who earn more than the average salary in the company.

Logic Explanation:

1. First, calculate the average salary using the AVG() function.
2. Use a WHERE clause to filter employees whose salary is greater than the calculated average salary.
3. Use SELECT to display the employee name and salary.

Scenario 2: Customer Orders without Matching Records

Question:

Retrieve a list of customer names who have not placed any orders.

Logic Explanation:

1. Perform a LEFT JOIN between the customers table and the orders table using customer_id as a common field.
2. After the join, filter records using WHERE orders.customer_id IS NULL to find customers without any matching orders.
3. Select the customer_name from the customers table.

Scenario 3: Product Sales Summary

Question:

Display the total sales amount for each product.

Logic Explanation:

1. Perform an INNER JOIN between the products table and the sales table using product_id.
2. Use the SUM() function to calculate the total sales amount for each product.
3. Apply a GROUP BY clause on product_name to aggregate the sales data.
4. Display product_name and the total sales amount.

Scenario 4: Department-Wise Employee Count

Question:

List each department name with the number of employees working in it.

Logic Explanation:

1. Perform a LEFT JOIN between the employees table and the departments table using the department_id.
2. Use the COUNT() function to count the number of employees in each department.
3. Apply a GROUP BY on the department_name to aggregate the data.
4. Select the department name and the employee count.

Scenario 5: Top 3 Highest Sales**Question:**

Find the top 3 highest sales transactions.

Logic Explanation:

1. Select the necessary columns like sale_id, customer_id, and amount from the sales table.
2. Use the ORDER BY clause on the amount column in descending order to list the highest sales first.
3. Use LIMIT 3 to restrict the output to the top 3 records.

Scenario 6: Calculate Employee Salary Ranks by Department**Question:**

Write a query to display each employee's name, department name, salary, and their salary rank within their respective department.

Logic:

- Use the **RANK()** window function to rank employees based on their salary within each department.
- Perform an **INNER JOIN** to combine employee data with department information using department_id.
- Display employee details with their salary rank using the **ORDER BY** clause.